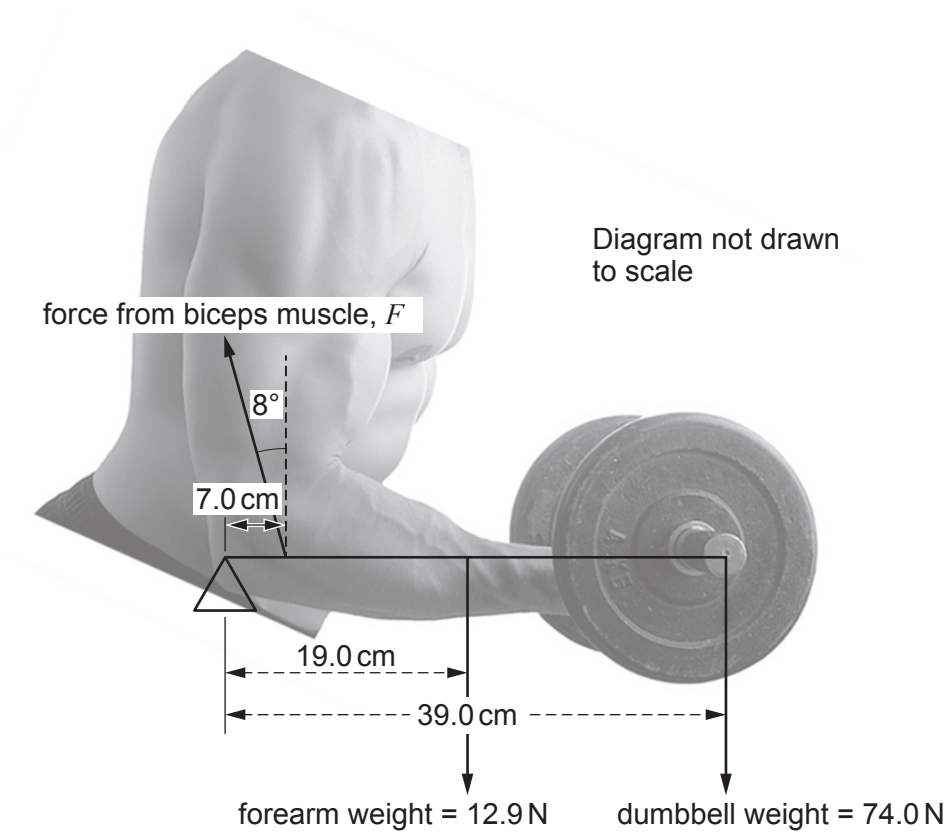


Option C – The Physics of Sports

9. (a) An athlete lifts a dumbbell of weight, 74.0 N as shown in the following diagram. Calculate the magnitude of the force, F , that the biceps muscle exerts on the athlete's forearm assuming that the elbow can be modelled as a pivot. [2]



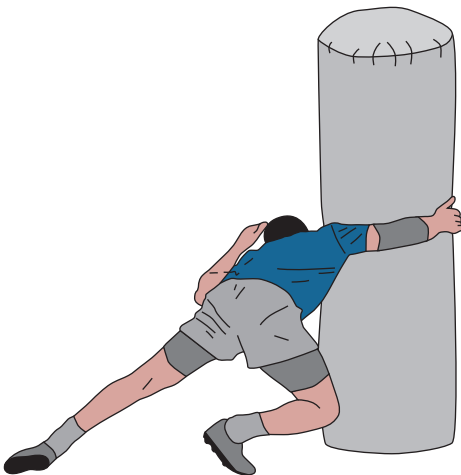
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- (b) A rugby player of mass 117 kg runs at a tackle pad with an initial velocity 7.75 m s^{-1} . In the collision with the tackle pad, the player experiences a mean force of 5500 N for a contact time of 0.176 s. This force is in the exact opposite direction to the initial velocity. Calculate the final velocity of the rugby player after the collision. [3]



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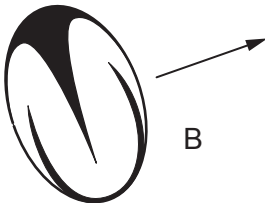
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- (c) (i) During a rugby match, a player wishes to kick the ball as far as possible. Describe the factors that affect the drag force on the ball. You may use the following diagrams to support your answer. [3]



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- (ii) A kicking coach thinks that, if the number of revolutions per second of the ball is doubled, then the rotational kinetic energy of a rugby ball will also double. Evaluate whether or not the coach is correct. [2]

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- (iii) The rugby ball is dropped on to two playing surfaces, A and B. Evaluate whether the playing surfaces are different. [3]

Playing surface	Initial height/m	Bounce height/m
A	7.8	1.2
B	4.5	0.7

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- (iv) When a rugby ball is kicked it gains an angular velocity of 37.7 rad s^{-1} in a time of 17.0 ms . The moment of inertia of the rugby ball is 0.0028 kg m^2 . Calculate the torque exerted on the ball by the kicker's foot. [2]
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- (v) A player throws a ball from a height of 2.0 m with an initial speed of 12 ms^{-1} at an angle of 32° to the horizontal. The maximum height reached by the player catching the ball is 4.4 m and he is standing 6.2 m away from the thrower. Explain why it is possible for the catcher to catch the ball. [5]

